

By generalization, I started to label a rare event as any behavior where the adage "beware of calm waters" can hold. Popular wisdom often warns of the old neighbor who appears to remain courtly and reserved, the model of an excellent citizen, until you see his picture in the national paper as a deranged killer who went on a rampage. Until then, he was not known to have committed any transgression. There was no way to predict that such pathological behavior could emanate from such a nice person. I associate rare events with any misunderstanding of the risks derived from a narrow interpretation of past time series.

Rare events are always unexpected, otherwise they would not occur. The typical case is as follows. You invest in a hedge fund that enjoys stable returns and no volatility, until one day, you receive a letter starting with "An unforeseen and *unexpected* event, deemed a rare occurrence ..." (emphasis mine). But rare events exist precisely because they are unexpected. They are generally caused by panics, themselves the results of liquidations (the investors rushing to the door simultaneously by dumping anything they can put their hands on as fast as possible). If the fund manager or trader expected it, he and his like-minded peers would not have invested in it, and the rare event would not have taken place.

The rare event is not limited to one security. It can readily affect the performance of a portfolio. For example, many traders engage in the purchase of mortgage securities and hedge them in some manner to *offset the risks and eliminate the volatility, hoping to derive some profits* in excess of the Treasury bond returns (which is used as the benchmark of the minimum expected returns on an investment). They use computer programs and draw meaningful assistance from Ph.D.s in applied mathematics, astrophysics, particle physics, electrical engineering, fluid dynamics, or sometimes (though rarely) plain Ph.D.s in Finance. Such a portfolio shows stable returns for long periods. Then, suddenly, as if by accident (I consider that *no* accident), the portfolio drops by 40% of its value when you expect, at the worst, a 4% drop. You call the manager to express your anger and he tells you that it was not his fault, but somehow the relationship dramatically changed (literally). He will also point out to you that similar funds also experienced the same problems.

Recall that some economists call the rare event a "peso problem". The designation peso problem does not appear to be undeservedly

stereotypical. Things have not gotten better since the early 1980s with the currency of the U.S.'s southern neighbor. Long periods of stability draw hordes of bank currency traders and hedge fund operators to the calm waters of the Mexican peso; they enjoy owning the currency because of the high interest rate it commands. Then they "unexpectedly" blow up, lose money for investors, lose their jobs and switch careers. Then a new period of stability sets in. New currency traders come in with no memory of the bad event. They are drawn to the Mexican peso, and the story repeats itself.

It is an oddity that most fixed income financial instruments present rare events. In the spring of 1998, I spent two hours explaining to a then-important hedge fund operator the notion of the *peso problem*. I went to great lengths to explain to him that the concept was generalized to every form of investment that was based on a naïve interpretation of the volatility of past time series. The reply was: "You are perfectly right. We do not touch the Mexican peso. We only invest in the Russian ruble". He blew up a few months later. Until then, the Russian ruble carried attractive interest rates, which invited yield hogs of all types to get involved. He and other holders of investments denominated in rubles lost close to 97% of their investment during the summer of 1998.

We saw in Chapter 3 that the dentist does not like volatility as it causes a high incidence of negative pangs. The closer he observes his performance, the more pain he will experience owing to the greater variability at a higher resolution. Accordingly investors, merely for emotional reasons, will be drawn into strategies that experience *rare but large* variations. It is called pushing randomness under the rug.

We can look at other aspects of the problem; think of someone involved in scientific research. Day after day, he will engage in dissecting mice in his laboratory, away from the rest of the world. He could try and try for years and years without anything to show for it. His significant other might lose patience with the *loser* who comes home every night smelling of mice urine. Until bingo, one day he comes up with a result. Someone observing the time series of his occupation would see absolutely no gain, while every day would bring him closer *in probability* to the end result.

The same with publishers; they can publish dog after dog without their business model being the least questionable, if once every decade